



Mass and Weight

The quantity of matter in an object is called its **mass**.

In our daily life, we use the word '**weight**' instead of 'mass'.

Thus, the weight of an object tells us how much of an object is there or how heavy an object is.

Units of Measuring Weight

The standard unit of weight is **kilogram**.

The smaller unit of weight is **gram**.

A still smaller unit of weight is **milligram**.

In short, we denote kilogram by **kg**, gram by **g** and milligram by **mg**.

As we have studied earlier,

1. we use the unit **milligram (mg)** to measure very small quantities of things like medicine, gold etc.
2. we use the unit **gram (g)** to measure light weights such as one or two apples, a book, some spices etc.
3. we use the unit **kilogram (kg)** to measure heavy weights such as a man, a bag of wheat or sugar, a bag of fruits or vegetables etc.

The relation between kilogram and gram is given by:

$$1 \text{ kg} = 1000 \text{ g}$$

Devices to Measure Weight

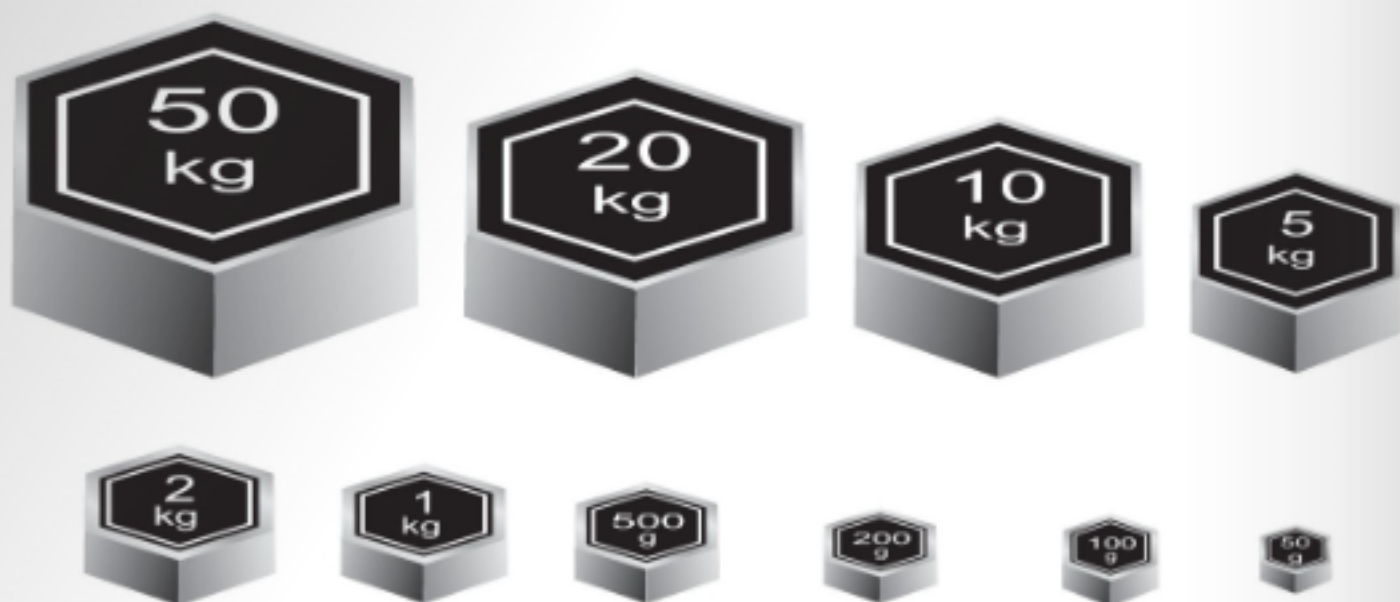
The most commonly used device to measure weight is a physical balance.

A balance has two pans.

We put the object in one pan of the balance and go on putting standard weights in the other pan till the two pans are at the same level. The total measure of these weights gives the actual weight of the object.



In general, weights of the following denominations are used:



Exercise 47

1. Write the suitable unit (kg or g) for measuring the weight of:

(a) yourself

(b) a tomato

(c) a pumpkin

(d) your maths book

(e) your school bag

(f) a loaf of bread

(g) a cricket ball

(h) a chocolate bar

2. Estimate for which of the following you need a bigger bag to carry:

(a) 1 kg potatoes or 1 kg potato chips

.....

(b) 1 kg salt or 1 kg popcorn

.....

(c) 1 kg groundnuts or 1 kg sugar

.....

(d) 1 kg peas or 1 kg cauliflower

.....



Addition and Subtraction of Weights

On the basis of the above principle, we have the following rules:

Rule 1: To add two or more weights in grams or kilograms, we simply add the numbers and put down the unit.

Observe the following additions.

			1	← Carry
	3	6	kg	
+	4	8	kg	
	8	4	kg	

			1	1	← Carry
	1	6	3	kg	
	3	0	8	kg	
+	2	9	6	kg	
	7	6	7	kg	

			1	1	← Carry
	5	6	7	kg	
+	3	8	9	kg	
	9	5	6	kg	

Rule 2: To subtract one weight from the other, we simply subtract the numbers and put down the unit.

Observe the following subtractions.

			8	12	← After borrowing
	9	2	kg		
-	6	8	kg		
	2	4	kg		

			5	13	← After borrowing
	6	5	kg		
-	3	7	kg		
	2	8	kg		

				12	← After borrowing
	8	2	10	kg	
	9	3	0	kg	
-	6	7	5	kg	
	2	5	5	kg	

Relation between kg and g

Repeat the above activity with two bags each measuring 500 g.

You shall observe: $500\text{ g} + 500\text{ g} = 1\text{ kg}$

But adding as shown above, we have

	5	0	0	g
+	5	0	0	g
	10	0	0	g

So, $1\text{ kg} = 1000\text{ g}$.



Conversion of Kilograms into Grams and Vice Versa

We know that : $1 \text{ kg} = 1000 \text{ g}$

Rule 1: To convert kilograms into grams, we multiply the number of kilograms by 1000.

Thus, $4 \text{ kg} = (4 \times 1000) \text{ g} = 4000 \text{ g}$;

$9 \text{ kg} = (9 \times 1000) \text{ g} = 9000 \text{ g}$.

Rule 2: To convert kilograms and grams into grams, we multiply the number of kilograms by 1000 and add it to the number of grams.

Thus, $5 \text{ kg } 675 \text{ g} = (5 \times 1000) \text{ g} + 675 \text{ g}$
 $= 5000 \text{ g} + 675 \text{ g} = 5675 \text{ g}$.

Example 1: Convert the following into grams.

(a) 8 kg

(b) $9 \text{ kg } 460 \text{ g}$

(c) $3 \text{ kg } 65 \text{ g}$

(d) $2 \text{ kg } 9 \text{ g}$

Solution: We have:

(a) $8 \text{ kg} = (8 \times 1000) \text{ g} = 8000 \text{ g}$.

(b) $9 \text{ kg } 460 \text{ g} = (9 \times 1000) \text{ g} + 460 \text{ g}$
 $= 9000 \text{ g} + 460 \text{ g}$
 $= 9460 \text{ g}$.

(c) $3 \text{ kg } 65 \text{ g} = (3 \times 1000) \text{ g} + 65 \text{ g}$
 $= 3000 \text{ g} + 65 \text{ g}$
 $= 3065 \text{ g}$.

(d) $2 \text{ kg } 9 \text{ g} = (2 \times 1000) \text{ g} + 9 \text{ g}$
 $= 2000 \text{ g} + 9 \text{ g}$
 $= 2009 \text{ g}$.



Example 2: Convert the following into kilograms and grams.

(a) 4685 g

(b) 7056 g

(c) 9005 g

Solution: We have:

(a) $4685 \text{ g} = 4000 \text{ g} + 685 \text{ g}$
 $= 4 \text{ kg} + 685 \text{ g}$
 $= 4 \text{ kg } 685 \text{ g}$.

(b) $7056 \text{ g} = 7000 \text{ g} + 56 \text{ g}$
 $= 7 \text{ kg} + 56 \text{ g}$
 $= 7 \text{ kg } 56 \text{ g}$.

(c) $9005 \text{ g} = 9000 \text{ g} + 5 \text{ g}$
 $= 9 \text{ kg} + 5 \text{ g}$
 $= 9 \text{ kg } 5 \text{ g}$.





Exercise 48

1. Convert the following into grams.

- (a) 5 kg (b) 6 kg 684 g (c) 2 kg 400 g
 (d) 3 kg 90 g (e) 8 kg 28 g (f) 1 kg 1 g

2. Convert the following into kilograms and grams.

- (a) 4000 g (b) 5764 g (c) 1085 g
 (d) 5006 g (e) 3050 g (f) 9875 g

Addition of Weights

Suppose two or more measures of weight (in kg and g) are given to us. To add these measures, we arrange them in two columns of kg and g and add them separately, as shown below.



Solved Examples



Example 1: Add 28 kg 460 g and 45 kg 385 g.

Solution: We may arrange and add the given measures columnwise, as under:

	kg		g
①	28	①	460
+	45		385
	73		845

$\therefore$ Sum = 73 kg 845

Step 1. Adding grams.

$$460 \text{ g} + 385 \text{ g} = 845 \text{ g.}$$

Write 845 under g column.

Step 2. Adding kg.

$$28 \text{ kg} + 45 \text{ kg} = 73 \text{ kg}$$

Write 73 under kg column.

In some cases, the sum of the figures in g column turns out to be greater than a multiple of 1000. In such cases, we convert as many grams to kg as possible and carry it over to the kg column. Consider the following example.

Example 2: Add 37 kg 875 g and 49 kg 387 g.

Solution: Arranging the given measures in two columns of kg and g and adding separately, we get:

kg			g		
1	1		1	1	
3	7		8	7	5
+	4	9	3	8	7
8			2		

Step 1. Adding grams.

$$\begin{aligned}
 875 \text{ g} + 387 \text{ g} &= 1262 \text{ g} \\
 &= 1000 \text{ g} + 262 \text{ g} \\
 &= 1 \text{ kg} + 262 \text{ g}.
 \end{aligned}$$

Write 262 under g column and carry over 1 to kg column.

Step 2. Adding kg.

$$1 \text{ kg (carried over)} + 37 \text{ kg} + 49 \text{ kg} = 87 \text{ kg}.$$

Write 87 under kg column.

$$\therefore \text{Sum} = 87 \text{ kg } 262 \text{ g}.$$

Short Cut Method:

Write the number of grams as 3-digit numbers.

For example, write 63 g as 063 g and 5 g as 005 g.

Expressing grams as 3-digit numbers, we may add such measures as ordinary numbers.

The following example will make the idea more clear.

Example 3: Add 34 kg 980 g, 56 kg 75 g and 9 kg 8 g.

Solution: Arrange the given measures in two columns of kg and g. Write the number of grams as 3-digit numbers. Now, add as ordinary numbers, as shown below.

kg			g		
2	1		1	1	
3	4		9	8	0
5	6		0	7	5
+		9	0	0	8
1			0		

$$\therefore \text{Sum} = 100 \text{ kg } 063 \text{ g}.$$





Exercise 49

Add the following.

1.

1	3	6	kg
+	7	9	kg

2.

5	4	8	kg	
+	2	7	4	kg

3.

7	4	5	g	
+	1	7	8	g

4.

kg	g
9	6 3 5
+	4 2 5 3

5.

kg	g
1 2	8 0 3
+	1 4 1 7 9

6.

kg	g
1 8	5 6 5
+	7 3 7 5

7.

kg	g
5 4	8 4 5
+	3 7 4 8 8

8.

kg	g
7 0	9 9 5
+	8 0 0 0 5

9.

kg	g
6 3	9 5 0
+	5 7 6 7 5

10.

kg	g
7 8	9 6 0
+	5 9 0 7 5

11.

kg	g
2 6	3 6 5
+	3 9 6 8 0

12.

kg	g
5 0 3	9 7 0
+	3 6 7 9 8 0

13.

kg	g
5 6	9 0 4
	3 8 4 1 7
+	1 1 2 5 6 8

14.

kg	g
6 5	3 9 4
	9 6 7 8
+	8 3 4 6 7

15.

kg	g
2 6 8	7 6 0
1 4 2	6 5 0
+	7 9 4 8 0

16.

kg	g
3 2 1	4 2 0
1 0 9	3 1 0
+	6 4 7 8 9



Arrange in columns and add.

17. 76 kg 706 g, 109 kg 89 g and 7 kg 7g
18. 13 kg 576 g, 46 kg 695 g and 2 kg 80 g
19. 108 kg 985 g, 15 kg 85 g and 6 kg 960 g
20. 825 kg 975 g, 245 kg 75 g and 603 kg 80 g



Subtraction of Weights

Suppose two measures of weight (in kg and g) are given to us. To subtract the smaller measure from the larger one, we arrange them in two columns of kg and g and subtract them separately, as shown below.

Example 1: Subtract 25 kg 375 g from 40 kg 400 g.

Solution : Arranging the given measures in two columns of kg and g and subtracting columnwise, we get:

kg		g		
3	10	3	10	10
4	0	4	0	0
-	2	5	3	7
	1	5	0	2

Step 1. Subtracting grams.

$$400 \text{ g} - 375 \text{ g} = 025 \text{ g}$$

Write 025 under g column.

Step 2. Subtracting kg.

$$40 \text{ kg} - 25 \text{ kg} = 15 \text{ kg}$$

Write 15 under kg column.

$$\therefore 40 \text{ kg } 400 \text{ g} - 25 \text{ kg } 375 \text{ g} = 15 \text{ kg } 25 \text{ g}.$$

Example 2: Subtract 18 kg 675 g from 62 kg 140 g.

Solution: Arranging the given measures in two columns of kg and g and subtracting columnwise, we get:

kg		g		
5	11	10	13	10
6	2	1	4	0
-	1	8	6	7
	4	3	4	6

Step 1. Subtracting grams.

We cannot subtract 675 g from 140 g.

So, we borrow 1 kg, leaving behind 61 kg.

Now, 1 kg 140 g = 1140 g.

$$\therefore 1140 \text{ g} - 675 \text{ g} = 465 \text{ g}.$$

Write 465 under g column.



Step 2. Subtracting kg:

$$61 \text{ kg} - 18 \text{ kg} = 43 \text{ kg}.$$

Write 43 under kg column.

$$\therefore 62 \text{ kg } 140 \text{ g} - 18 \text{ kg } 675 \text{ g} = 43 \text{ kg } 465 \text{ g}.$$

Short Cut Method:

Taking the number of grams as 3-digit numbers, arrange the given measures in two columns of kg and g. Now, subtract them as ordinary numbers, as shown below.

Example 3: Subtract 107 kg 375 g from 304 kg 50 g.

Solution : Taking the number of grams as 3-digit numbers, arrange the given measures in two columns of kg and g.

Now, subtract them as ordinary numbers, as shown below.

$$\therefore 304 \text{ kg } 50 \text{ g} - 107 \text{ kg } 375 \text{ g} = 196 \text{ kg } 675 \text{ g}.$$

kg	g
304	050
- 107	- 375
196	675



Exercise 50

Subtract:

1.

6	1	5	kg
-	4	6	7 kg

2.

5	3	0	g
-	3	7	4 g

3.

6	0	0	g
-	2	3	5 g

4.

kg	g
52	800
-	19485

5.

kg	g
73	423
-	38365

6.

kg	g
62	125
-	47496

7.

kg	g
403	320
-	304743

8.

kg	g
153	400
-	47575

9.

	kg		g		
	9	0	0	0	0
-	3	6	4	0	5

10.

	kg		g		
	3	1	5	0	7
-	1	8	7	1	8

11.

	kg		g		
	8	2	1	0	0
-	5	7	8	7	5

12.

	kg		g		
	5	0	0	0	0
-	1	4	9	1	0

13.

	kg		g		
	6		1	0	2
-	5		3	7	8

14.

	kg		g		
	1	0	0	0	0
-		0	3	7	5

15.

	kg		g		
	8		0	2	5
-	6		0	7	8

16.

	kg		g		
	1	8	2	1	0
-		9	6	2	5

Arrange in columns and find the difference between:

17. 58 kg 348 g and 9 kg 375 g

18. 70 kg 235 g and 58 kg 348 g

19. 10 kg 5 g and 8 kg 28 g

20. 2 kg and 385 g

21. 300 kg 50 g and 105 kg 765 g

22. 100 kg and 56 kg 125 g

Word Problems

Example 1: Mohan bought 3 kg 500 g of mangoes, 2 kg 375 g of oranges and 1 kg 50 g of apples.
What is the total weight of the fruits bought by him?

Solution : Weight of mangoes = 3 kg 500 g

Weight of oranges = 2 kg 375 g

Weight of apples = 1 kg 50 g

Total weight = 3 kg 500 g + 2 kg 375 g + 1 kg 50 g
= 6 kg 925 g.

Hence, the total weight of the fruits is 6 kg 925 g.

	kg	g		
		①		
	3	5	0	0
	2	3	7	5
+	1	0	5	0
	6	9	2	5

Example 2: A bucket full of water weighs 14 kg 320 g. The weight of the empty bucket is 1 kg 685 g. What is the weight of water in the bucket?

Solution : Weight of bucket full of water = 14 kg 320 g.

Weight of the empty bucket = 1 kg 685 g.

Weight of water in the bucket

$$= 14 \text{ kg } 320 \text{ g} - 1 \text{ kg } 685 \text{ g}$$

$$= 12 \text{ kg } 635 \text{ g.}$$

Hence, the weight of water in the bucket is 12 kg 635 g.

kg	g
14	320
- 1	685
13	635

Example 3: A grocer had 100 kg of sugar. He sold 15 kg 675 g to one customer and 32 kg 750 g to another customer. Find the weight of sugar left with him.

Solution : Total quantity of sugar with the grocer = 100 kg

Quantity of sugar sold to one customer = 15 kg 675 g

Quantity of sugar sold to another customer = 32 kg 750 g.

Total quantity of sugar sold = 15 kg 675 g + 32 kg 750 g

$$= 48 \text{ kg } 425 \text{ g.}$$

Quantity of sugar left with him = 100 kg - 48 kg 425 g

$$= 51 \text{ kg } 575 \text{ g.}$$

Hence, the quantity of sugar left with the grocer is 51 kg 575 g.

kg	g
100	
- 15	675
- 32	750
48	425

kg	g
100	
- 48	425
51	575



Exercise 51

1. A packet of sweets weighs 2 kg 375 g. Another packet weighs 1 kg 950 g. How much do they weigh together?
2. Rajni bought 16 kg 350 g of rice from one shop and 13 kg 750 g of rice from another shop. How much rice did she buy in all?
3. Rahul bought 24 kg 850 g of rice; 7 kg 750 g of wheat and 4 kg 350 g of ragi for a month. What is the total weight of grains he bought?
4. Nirmal bought 2 kg 750 g of potatoes; 1 kg 880 g of tomatoes; 2 kg 375 g of brinjal and 940 g of lady finger. What is the total weight of vegetables bought by him?

5. Anil weighs 38 kg 485 g. His brother Sumeet is heavier than him by 5 kg 755 g. How much does Sumeet weigh?
6. A tin contains 17 kg 285 g of refined oil. The empty tin weighs 1 kg 365 g. What is the total weight of the tin with refined oil?
7. Sonu weighs 36 kg 200 g while his sister Monica weighs 27 kg 765 g. Who weighs more and by how much?
8. A tin full of biscuits weighs 10 kg 50 g and the empty tin weighs 1 kg 225 g. What is the weight of the biscuits in the tin?
9. Renu weighed 30 kg 360 g. She fell ill and lost 3 kg 485 g. What is her weight now?
10. The total weight of Rita and Geeta is 77 kg 340 g. If Rita weighs 38 kg 765 g, how much does Geeta weigh?
11. On Diwali festival Maneka bought 10 kg of sweets. She distributed 7 kg 275 g among her friends. How much sweets were left with her?
12. Preeti bought 12 kg 300 g of sugar in the beginning of the month. She used 11 kg 725 g of it during the month. How much sugar is left unused?
13. The total weight of a bag containing three packets of rice, sugar and tea was 17 kg. If the weight of the rice was 9 kg 650 g and the weight of sugar was 6 kg 475 g, find the weight of the packet containing tea.
14. The total weight of three girls is 95 kg. If two of them weigh 27 kg 325 g and 32 kg 775 g respectively, what is the weight of the third girl?



Things to Remember

1. The quantity of matter in an object is called its mass.
2. In our daily life, we use the word weight instead of mass.
3. We measure weights in grams and kilograms.
4. In short, we write kilograms as kg and grams as g.
5. We write the number of grams as 3-digit numbers.
So, we write 75 g as 075 g and 5 g as 005 g.
6. To add the measures in kg and g, write the measures in two columns of kg and g. Write the number of grams as 3-digit numbers. Now, add as ordinary numbers.
7. To subtract the measures in kg and g, write them in column form and subtract as ordinary numbers.



**QUESTION BAG 1****(Objective Type Questions)****Tick (✓) the correct answer.**

1. Which of the following would you use to measure the weight of a rock?

(a) Scale ☐ (b) Balance ☐
(c) Compass ☐ (d) Thermometer ☐

2. Which of the following units would you use to measure the weight of a bicycle?

(a) gram ☐ (b) milligram ☐ (c) kilogram ☐ (d) miles ☐

3. Which of the following is not equal to 1 kg?

(a)     ☐

(b)      ☐

(c)   ☐

(d)        ☐

4. If 5 apples together weigh 1 kg, the weight of each apple is

(a) 100 g ☐ (b) 200 g ☐ (c) 250 g ☐ (d) 500 g ☐

5. 10 apples weigh as much as 2 watermelons. How many watermelons would weigh the same as 25 apples?

(a) 3 ☐ (b) 4 ☐ (c) 5 ☐ (d) 6 ☐

6. If 3 potatoes together weigh 450 g, what is the weight of each potato?

(a) 100 g ☐ (b) 150 g ☐ (c) 200 g ☐ (d) 250 g ☐

7. Which of the following would weigh closest to 70 kg?

(a) A butterfly ☐ (b) A dog ☐ (c) A cow ☐ (d) A car ☐

8. 3 kg 60 g is equal to

(a) 3600 g ☐ (b) 3006 g ☐ (c) 3060 g ☐ (d) 3660 g ☐

9. There are two bags of rice, A and B. B is heavier than A and by how much?



- (a) 3 kg 660 g ☐ (b) 3 kg 440 g ☐ (c) 4 kg 560 g ☐ (d) 4 kg 760 g ☐
10. $2 \text{ kg} - 280 \text{ g} = \dots\dots\dots \text{ g}$
 (a) 1720 ☐ (b) 1820 ☐ (c) 1780 ☐ (d) 2720 ☐
11. $5 \text{ kg} - 3 \text{ kg } 375 \text{ g} = ?$
 (a) 1 kg 625 g ☐ (b) 2 kg 625 g ☐ (c) 1 kg 650 g ☐ (d) 2 kg 650 g ☐
12. $3675 \text{ g} - 2 \text{ kg } 550 \text{ g} = \dots\dots\dots \text{ g}$
 (a) 125 ☐ (b) 1025 ☐ (c) 1125 ☐ (d) 1225 ☐

QUESTION BAG 2

1. Write the suitable unit (kg or g) to measure the weight of:

- | | |
|---|---|
| (a) your pencil ☐ | (b) a bag of rice ☐ |
| (c) a shoe ☐ | (d) an office table ☐ |
| (e) your pencil box ☐ | (f) a television set ☐ |
| (g) a pile of 40 notebooks ☐ | |

2. The addends in each problem are the same. Find the addend in each case.

- (a) g = 1 kg
- (b) g + g = 1 kg
- (c) g + g + g + g = 1 kg

3. Circle the best estimate.

- | | | | |
|-----------------------------------|-------|--------|---------|
| (a) A car weighs about: | 50 kg | 100 kg | 1000 kg |
| (b) A mobile phone weighs about: | 10 g | 100 g | 1 kg |
| (c) A birthday cake weighs about: | 50 g | 200 g | 2 kg |



(d) An inland letter weighs about: 10 g 50 g 100 g

(e) A new-born baby weighs about: 500 g 3 kg 10 kg

4. Fill in the empty boxes.

(a) $500 \text{ g} + \boxed{} \text{ g} = 1 \text{ kg}$

(b) $800 \text{ g} + \boxed{} \text{ g} = 1 \text{ kg}$

(c) $600 \text{ g} + \boxed{} \text{ g} = 1 \text{ kg}$

(d) $300 \text{ g} + \boxed{} \text{ g} = 1 \text{ kg}$

(e) $150 \text{ g} + 200 \text{ g} + \boxed{} \text{ g} = 1 \text{ kg}$

5. Put the correct sign <, > or = in the boxes.

(a) $2 \text{ kg } 24 \text{ g} \boxed{} 2224 \text{ g}$

(b) $1 \text{ kg } 500 \text{ g} \boxed{} 1500 \text{ g}$

(c) $6 \text{ kg } 600 \text{ g} \boxed{} 6060 \text{ g}$

(d) $5 \text{ kg } 405 \text{ g} \boxed{} 5045 \text{ g}$

(e) $7 \text{ kg } 98 \text{ g} \boxed{} 7980 \text{ g}$

6. At a railway station, a coolie carried a suitcase weighing 26 kg 756 g and a bag weighing 19 kg 554 g. Find the total weight carried by the coolie.
7. For a dinner party, 15 kg 485 g of wheat, 9 kg 675 g of rice and 8 kg 955 g of sugar were purchased. Find the total weight of the goods purchased.
8. A lady who weighed 84 kg 336 g joined a weight loss clinic. As a result she lost 8 kg 678 g. How much does she weigh now?

